

Digital Image Processing

Assignment 1: Down Sampling and Up Sampling

1. Objective

This experiment is an evaluation of image resizing. For down sampling, three methods are used in this experiment: Max, Min, and Mean. Meanwhile, three methods are also used for up sampling: Nearest Neighbor, Bilinear, and Bicubic.

2. Down Sampling Analysis

In this experiment, down sampling is used to reduce the resolution to $1/5$ of the original size. Therefore, the pixels of the original image are grouped into 5×5 blocks.

Based on the results, the Max method produces an image that appears brighter than the others. This occurs because the Max method takes the highest intensity value from each matrix block. However, because it takes the highest intensity value, the resulting image may appear to have more noise compared to the other methods.

Meanwhile, the Min method produces an image that appears darker than the others, considering that this method takes the lowest intensity value from each block, making it the opposite of the Max method.

Finally, the Mean method produces the most balanced result because it uses the average value of all pixels within each block. Therefore, the resulting image appears smoother.

3. Up Sampling Analysis

This experiment also performs up sampling using three methods: Nearest Neighbor, Bilinear, and Bicubic. The first method is Nearest Neighbor. This method uses the coordinates of the pixels in an image to determine the intensity value to be used. The pixel from the nearest neighboring position is selected based on the position being calculated. Therefore, the resulting image appears pixelated.

Next, Bilinear is different from Nearest Neighbor. Bilinear does not rely on only one pixel, but instead uses four neighboring pixels. These four pixels are interpolated horizontally and vertically while considering the distance of each pixel from the position being calculated. Therefore, the resulting image appears smoother and less pixelated compared to Nearest Neighbor.

The final method is Bicubic. Bicubic uses a similar concept, but instead of using one or four neighboring pixels, it uses the surrounding pixels in a 4×4 area, or 16 pixels, around the position being calculated. These pixels are then used for cubic interpolation in the horizontal and vertical directions. With more pixel information being used, this method generally produces a smoother result.